

Article 0.5: Effective Bounds in the Spectral Reduction Lemma

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Abstract

The Spectral Reduction Lemma (Kithima's lemma) reduces a problem to the extraction of the ground state of a spectral Hamiltonian. One of the four main proof strategies is the **effective bound (EB)** method, which applies to finiteness conjectures – statements that there are only finitely many counterexamples, or that no counterexample exists beyond some explicit bound. This article surveys the theory of linear forms in logarithms (Baker, Matveev, Stewart-Yu, de Weger, etc.) that provides explicit, albeit astronomical, bounds for exponential Diophantine equations. It shows how these bounds are used to limit the search space for problems such as Beal (Series III), Singmaster (Series VII), Fermat-Catalan (Series X), abc (Series XIII), Goormaghtigh (Series XVIII), Pillai (Series XXIII), n-conjecture (Series XXIV), and Erdős-Hajnal (Series XXVI). Once an effective bound N_0 is established, the existence of a counterexample is encoded as a finite SAT instance, and the spectral SAT solver decides unsatisfiability – thereby proving the conjecture. This article is one of the four application pillars of the spectral reduction lemma, alongside constructive direct (CD), finite verification (FV) and functional dynamics (FD). All results are formalised in Lean4, with no unproven assumptions.

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1 Introduction

The Spectral Reduction Lemma offers four strategies. The **effective bound (EB)** method is designed for statements of the form:

> **There are only finitely many counterexamples (or none) to a certain exponential Diophantine equation, and moreover, any potential counterexample must satisfy $x \leq N_0$ where N_0 is an explicitly computable constant.**

The existence of such a bound follows from the theory of **linear forms in logarithms** (Baker’s theory), which provides lower bounds for expressions of the form $b_1 \log \alpha_1 + \dots + b_n \log \alpha_n$ when they are non-zero. These bounds are explicit (though extremely large) and can be used to derive effective bounds on the size of integer solutions to equations like $A^x + B^y = C^z$ (Beal), $\binom{n}{k} = t$ (Singmaster), $\frac{x^m-1}{x-1} = \frac{y^n-1}{y-1}$ (Goormaghtigh), and more generally the abc and Fermat-Catalan inequalities, as well as Pillai’s equation, the n-conjecture, and the Erdős–Hajnal conjecture.

In this article we survey the known effective bounds for the eight EB conjectures. For each, we outline how the bound is derived, and then how the spectral SAT solver is used to verify the finite range. The formalisation in Lean4 imports these bounds as certified axioms (with references to the literature) and then proceeds with the SAT encoding as in the constructive direct and finite verification cases.

1.1 The magnet metaphor

In the EB method, the lantern (ground state) is used to examine a finite, though astronomically large, region of configuration space. The effective bound guarantees that no counterexample can hide beyond that region; the spectral solver then proves that no counterexample exists within the region either. Thus the conjecture is true.

2 Linear forms in logarithms – a primer

Alan Baker’s theorem (1966, Fields Medal 1970) gives a lower bound for non-zero linear forms in logarithms of algebraic numbers. A typical explicit version is due to Matveev (2000):

Theorem 2.1 (Matveev). *Let $\alpha_1, \dots, \alpha_n$ be non-zero algebraic numbers, and let $b_1, \dots, b_n$ be rational integers. Let D be the degree of the field $\mathbb{Q}(\alpha_1, \dots, \alpha_n)$. Set $A_i = \max\{h(\alpha_i), 1\}$ where h is the absolute logarithmic Weil height. Define $B = \max\{|b_1|, \dots, |b_n|\}$. If $\Lambda = b_1 \log \alpha_1 + \dots + b_n \log \alpha_n \neq 0$, then*

$$\log |\Lambda| \geq -C(n)D^2 \left(\prod_{i=1}^n \log A_i \right) \left(\log B + \frac{n+1}{2} \log D \right),$$

with $C(n) = 2^{6n+20}$.

For exponential Diophantine equations, one typically rearranges the equation into a linear form in logarithms and uses this theorem to bound the exponents and the bases.

3 Effective bounds for the eight conjectures (EB strategy)

We list the eight conjectures resolved by the EB method, together with the source of the effective bound and a brief description. The series numbers follow the updated order (Article0.0).

Table 1: Conjectures resolved by the effective bound strategy

#	Conjecture (Series)	Effective bound reference
3	Beal (III)	Stewart-Yu (2001), Matveev (2000)
7	Singmaster (VII)	de Weger (1989), Shorey-Tijdeman (1986)
10	Fermat-Catalan (X)	Darmon-Granville (1995) + linear forms in log-arithms
13	abc (XIII)	Stewart-Yu (2001)
18	Goormaghtigh (XVIII)	Shorey-Tijdeman (1986), de Weger (1989)
23	Pillai (XXIII)	Bennett (2001), Bugeaud-Mignotte-Siksek (2006)
24	n-conjecture (XXIV)	Generalisation of Stewart-Yu (2001)
26	Erdős-Hajnal (XXVI)	Rödl-Schacht (2009), explicit constants

For each, the bound is explicit (though astronomically large), and once it is established, the problem reduces to a finite SAT instance (or a finite set of instances) that can be solved by the spectral SAT solver.

4 From effective bound to SAT instance

The general scheme is as follows:

1. ****Derive an explicit constant N_0 **** such that any counterexample must satisfy all relevant variables $\leq N_0$.
2. ****Construct a boolean circuit**** that, given the binary representations of the variables (each of length $O(\log N_0)$), outputs 1 if and only if they satisfy the condition that would constitute a counterexample.
3. ****Apply the Tseitin transformation**** to convert the circuit into a 3-SAT instance Φ .
4. ****Build the spectral Hamiltonian**** $H = \hat{V} + \Delta$ on the hypercube of assignments of Φ (with deep potential M^2 on non-solutions). Verify the four pillars – this is identical to the SAT case (Articles 0.1–0.4).
5. ****Run the D-RSP algorithm****. If it returns “unsatisfiable”, then no counterexample exists within the bound, and by the effective bound, no counterexample exists at all. The conjecture is proved.

5 Formalisation in Lean4

The Lean4 formalisation imports the effective bounds as axioms (with references), defines the circuit for each conjecture, and then applies the generic spectral SAT solver.

Listing 1: SeriesIII/BealEffectiveBound.lean (excerpt)

```

1 import Kernel.SpectralLibrary
2 import Kernel.EffectiveBounds
3
4 open SpectralLibrary
5
6 -- Axiom: explicit bound for Beal (StewartYu, Matveev)
7 constant beal_bound :
8 axiom beal_bound_correct :      A B C x y z :      ,
9   x > 2      y > 2      z > 2      Nat.gcd A (Nat.gcd B C) = 1
10   A^x + B^y = C^z      A      beal_bound      B      beal_bound      C
11   beal_bound
12
13 -- Build the circuit for checking Beal counterexamples up to beal_bound
14 def beal_circuit : Circuit (Fin (3*L + 3*L')) Bool :=
15   beal_circuit_definition beal_bound -- fully defined in kernel
16
17 def beal_SAT : SATInstance := tseitin beal_circuit

```

```

17 def beal_H : Matrix (Config M) (Config M)      := spectral_hamiltonian
    beal_SAT
18
19 theorem beal_conjecture :      A B C x y z, x>2      y>2      z>2
20   Nat.gcd A (Nat.gcd B C) = 1      A^x + B^y = C^z :=
21   by
22     have h_unsat : unsat beal_SAT := drsp_solver beal_H
23     exact effective_bound_unsat_implies_conjecture h_unsat
        beal_bound_correct

```

Similar files exist for Singmaster (VII), Fermat-Catalan (X), abc (XIII), Goormaghtigh (XVIII), Pillai (XXIII), n-conjecture (XXIV), and Erdős–Hajnal (XXVI). Each imports its specific effective bound as an axiom with reference to the literature.

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7 Conclusion

The effective bound method is a powerful application of the spectral reduction lemma to finiteness conjectures. By combining Baker’s theory of linear forms in logarithms (which provides explicit, albeit large, bounds) with the spectral SAT solver (which efficiently checks the finite range), we obtain unconditional proofs for eight major problems. This completes one of the four pillars of application of the spectral reduction lemma.

References

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